

OPTIMAL SIZING OF LEVERAGED CRYPTO LONG–SHORT POSITIONS UNDER KOU JUMP-DIFFUSION VIA KILLED MOMENTS

We study a leveraged crypto long–short position, held on a DeFi lending protocol, whose solvency is governed by a collateral-to-liability health factor and whose payoff is killed when that factor crosses a liquidation boundary. Sizing such a position requires the survival probability and killed-payoff moments at every candidate initial buffer. We derive a semi-analytic Laplace-domain method for these quantities under bivariate Kou double-exponential jump–diffusion dynamics. Compared with Monte Carlo simulation, the method is free of sampling noise, stays accurate near the liquidation boundary where discretely monitored simulation is least reliable, and evaluates orders of magnitude faster, making repeated evaluation across candidate buffers practical. We use these outputs to solve the initial-buffer sizing problem under alternative liquidation constraints and moment-based objectives, and a paired moving-block bootstrap shows that the selected buffer is sensitive to calibration uncertainty.

Keywords: DeFi lending; crypto assets; leveraged long–short; killed payoff moments; health factor; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent

JEL Classification: G11; G12; C65

1. Introduction

Crypto-asset DeFi lending protocols such as AAVE and Compound enable a leveraged long–short strategy: an investor deposits one crypto asset as collateral, borrows a second crypto asset, and sells it short. The position is governed by a real-time *health factor*—a borrowing-power-adjusted ratio of collateral to liability—and may be liquidated when that ratio falls below one. The resulting exposure combines leveraged relative-value risk, a path-dependent liquidation boundary, discontinuous price jumps, and state-dependent survival. Its initial health buffer controls both leverage and distance to liquidation, but evaluating that trade-off first requires the distribution of a payoff killed at the liquidation boundary.

The two recurring objects are the probability that the position survives to the horizon and the moments of its killed terminal payoff, and both must be evaluated repeatedly across candidate initial health buffers. Monte Carlo simulation can approximate them, but barrier discretisation and sampling noise are especially consequential near liquidation, and repeated simulation is costly inside a sizing search. The missing input for solving the sizing problem is therefore not a universal economic objective, but a fast, repeatable map from the initial buffer to survival probabilities and killed-payoff moments.

We work under a bivariate Kou double-exponential jump–diffusion model (Kou, 2002) and make four contributions. First, a unit-equity normalization collapses the two-asset holding to a single initial health buffer, the natural sizing parameter, while liquidation depends only on a one-dimensional log-price ratio between the two legs. Second, for each admissible moment order, an exponential change of measure absorbs the short-leg price factor, reducing the two-asset killed moment to a one-dimensional killed

expectation of that log-price ratio. Third, the double-exponential jump structure turns the resulting Laplace-domain equation into algebra: in the generic case, three barrier conditions pin down a fixed, low-dimensional linear system whose size does not grow with the moment order; numerical Laplace inversion then recovers the time-domain quantities (Talbot, 1979). Fourth, these objective-independent outputs turn initial-buffer sizing into a one-dimensional optimization, which we illustrate under alternative liquidation constraints and moment-based objectives without taking a position on which objective is economically correct.

This paper sits at the intersection of the DeFi liquidation literature and the analytical first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations, but do not provide the repeated killed-moment evaluation developed here. Kou and Wang (2003) derive explicit Laplace-transform first-passage results under double-exponential jump diffusion, and Sepp (2004) extends this to barrier option pricing; we instead price a barrier-killed long–short ratio, bridging the two settings via the measure-change reduction above. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs, and Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our sizing problem is a static analogue, optimizing a single initial buffer against a killed-payoff distribution rather than a control process.

The scope of this framework is as follows. It evaluates liquidation probabilities, killed payoff moments, and conditional surviving-path moments repeatedly across initial health buffers, and solves the resulting sizing problem for a stated objective.

The remainder of the paper is organised as follows. Section 2 defines the position, the liquidation barrier, and the health-buffer parameterization. Section 3 specifies the bivariate Kou model and its tilted dynamics. Section 4 reduces payoff moments to one-dimensional killed expectations, and Section 5 derives their time-Laplace transforms and finite-dimensional evaluation system. Section 6 validates the numerical method, and Section 7 illustrates the initial-buffer sizing problem. Section 8 concludes. Appendix A describes the empirical calibration estimator, Appendix B derives the downward-phase boundary conditions, and Appendix C collects the proofs.

2. Leveraged Long–Short Position and Health-Factor Barrier

We consider a leveraged long–short portfolio in which an investor holds asset S_1 and owes asset S_2 , with net initial equity normalized to one unit of account. The position is liquidated if collateral becomes too small relative to the liability. This section defines the barrier, the killed-payoff object studied in the paper, and the mapping from the initial health buffer to holdings. No investor objective is needed for these definitions.

2.1. Health factor, liquidation time, and killed payoff

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \quad X_{i,0} = 0, \quad i = 1, 2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset S_1 (long leg) and $n_2 > 0$ units of asset S_2 (short leg).

Let

$$V_{1,t} := n_1 S_{1,t}, \quad V_{2,t} := n_2 S_{2,t}$$

denote the *collateral value* (long leg) and the *borrow value* (outstanding liability) at time t . Fix $b \in (0, 1)$, the *liquidation threshold* of asset S_1 (the Aave v3 parameter of the same name; distinct from

the maximum LTV at origination, which is a separate and typically lower protocol constant). Solvency requires $V_{2,t} \leq bV_{1,t}$; liquidation occurs when the borrow value exceeds adjusted collateral,

$$V_{2,t} > bV_{1,t}.$$

The *health factor* summarises this margin condition as a single ratio,

$$H_t := \frac{bV_{1,t}}{V_{2,t}} = \frac{bn_1S_{1,t}}{n_2S_{2,t}}, \quad h_t := \log H_t, \quad (1)$$

where h_t is the *log health factor*. The solvency condition is $H_t \geq 1$ ($h_t \geq 0$); breaching it is equivalent to the borrow value exceeding the b -adjusted collateral. Liquidation is triggered at the first such instant:

$$\tau := \inf\{t \geq 0 : H_t < 1\} = \inf\{t \geq 0 : h_t < 0\}.$$

A larger b permits a larger borrow-to-collateral ratio $V_2/V_1 \leq b$ before liquidation, giving the position more room to absorb adverse price moves; a smaller b is a stricter protocol threshold, triggering earlier forced exit and limiting bad-debt exposure.

Since

$$h_t = \log\left(\frac{bn_1S_{1,0}}{n_2S_{2,0}}\right) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\left(\frac{bn_1S_{1,0}}{n_2S_{2,0}}\right), \quad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : Z_t < -h_0\}.$$

The one-dimensional process Z thus carries all barrier-relevant risk: the position survives if and only if Z stays above the absorbing level $-h_0$. The parameter $h_0 = \log H_0$ is therefore the *initial log-cushion* — the log-distance of the initial health factor from the liquidation boundary $H = 1$. A larger h_0 means the position starts further from forced exit; $h_0 \rightarrow 0^+$ means the position starts almost at the boundary with negligible cushion.

The *killed terminal payoff* at horizon T is

$$\Pi_T := (n_1S_{1,T} - n_2S_{2,T}) \mathbf{1}_{\{\tau > T\}}.$$

On $\{\tau > T\}$ the health factor satisfies $H_T \geq 1$, so $bn_1S_{1,T} \geq n_2S_{2,T}$; since $b \in (0, 1)$ this forces $n_1S_{1,T} > n_2S_{2,T}$, meaning the position carries strictly positive terminal equity on every surviving path. No positive-part truncation is needed. The indicator $\mathbf{1}_{\{\tau > T\}}$ assigns zero to liquidated paths. This defines a *killed payoff*; it is a mathematical convention rather than a model of the investor's protocol-specific recovery. Since the initial net book value is normalised to one unit of account (Section 2.2), net profit on a surviving path is $R_T := \Pi_T - 1$, but all moments in this paper are stated in terms of Π_T directly. This paper characterises the killed moments of Π_T , the conditional distribution on $\{\tau > T\}$, and the probability of that event, with $H_t = 1$ treated as an absorbing boundary. Liquidation mechanics are outside the model scope.

Remark 1 (Recovery). When the boundary is crossed, a lending protocol initiates a protocol-specific, state-dependent liquidation procedure. Partial debt repayment, liquidation bonuses, keeper timing, market liquidity, and execution prices can all affect the value ultimately retained by the borrower (Qin et al.,

2021; Heimbach and Huang, 2024). A single-barrier first-passage model does not determine those cash flows.

We therefore set terminal wealth to zero following liquidation solely to define the killed-payoff benchmark Π_T . This convention is deliberately severe and analytically convenient; it is not intended to reproduce recovery cash flows or to assert that recovery has zero conditional mean. If R_τ denotes a specified recovery value carried to T , the corresponding investor-wealth object would instead be

$$W_T = (n_1 S_{1,T} - n_2 S_{2,T}) \mathbf{1}_{\{\tau > T\}} + R_\tau \mathbf{1}_{\{\tau \leq T\}} = \Pi_T + R_\tau \mathbf{1}_{\{\tau \leq T\}}.$$

The closed-form construction in this paper concerns the first term. Modelling the second requires additional protocol and execution assumptions.

Remark 2 (Dynamic collateral management). A practitioner who monitors the health factor could top up collateral or partially unwind the short leg to steer H_t away from the liquidation boundary. Such a strategy would replace the absorbing barrier with a reflecting or controlled boundary and the static evaluation problem with a stochastic control problem. We hold the position fixed to isolate the ex-ante killed-moment calculation; the assumption should not be interpreted as a claim that dynamic intervention is infeasible or suboptimal.

Thus killed payoff and investor terminal wealth are distinct objects. The former is the tractable object evaluated below; recovery and intervention overlays may change the latter and, consequently, any sizing decision built from it.

Using the definition of H_T , one obtains the following factorization.

Lemma 2.1 (Payoff factorization).

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} (e^{h_0 + Z_T} - b) \mathbf{1}_{\{\tau > T\}}. \quad (2)$$

The factor $(e^{h_0 + Z_T} - b)$ is strictly positive on $\{\tau > T\}$ because $H_T = e^{h_0 + Z_T} \geq 1 > b$, consistent with the definition of Π_T .

This separates the payoff into a numeraire factor driven by S_2 , and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0 .

2.2. Health-buffer parameterization and feasible set

The two coin holdings (n_1, n_2) represent two degrees of freedom. A normalization reduces this to a single scalar, and the barrier structure identifies h_0 as the natural evaluation parameter. This reduction does not itself select a portfolio.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1 + x)/x$. Solving for x ,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \quad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \quad (3)$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

This algebraic condition is necessary but not sufficient for a feasible DeFi position. Initial solvency additionally requires $H_0 = e^{h_0} > 1$, hence $h_0 > 0$. If the protocol imposes a maximum loan-to-value ratio $\ell_{\max} \in (0, b)$ at origination, then

$$\frac{V_{2,0}}{V_{1,0}} = \frac{b}{H_0} \leq \ell_{\max} \iff h_0 \geq h_0^{\min} := \log\left(\frac{b}{\ell_{\max}}\right) > 0.$$

Accordingly, define the feasible set

$$\mathcal{H}(b, \ell_{\max}) := \begin{cases} [\log(b/\ell_{\max}), \infty), & \text{when an origination LTV } \ell_{\max} \in (0, b) \text{ applies,} \\ (0, \infty), & \text{otherwise.} \end{cases} \quad (4)$$

The lower endpoint is included in the first case because the protocol permits origination at its maximum LTV. The point $h_0 = 0$ is excluded in the second case because it starts the position at the liquidation boundary. The initial gross leverage — long-leg value per unit of equity — follows directly from (3):

$$L_0 := n_1(h_0) S_{1,0} = \frac{e^{h_0}}{e^{h_0} - b}. \quad (5)$$

L_0 is strictly decreasing in h_0 . Algebraically, $L_0 \uparrow \infty$ as $h_0 \downarrow \ln b$, but this limit is infeasible because $\ln b < 0$ and the position would already be liquidatable at inception. Over the feasible set (4), maximum leverage is attained at the smallest permitted health buffer; as $h_0 \rightarrow \infty$, leverage $L_0 \downarrow 1$ (the short leg vanishes and the position becomes unlevered). Equivalently, the borrow-to-equity ratio is $L_0 - 1 = b/(e^{h_0} - b)$, which also collapses the initial health factor back to $H_0 = e^{h_0}$ via $H_0 = bL_0/(L_0 - 1)$. Choosing h_0 is therefore equivalent to choosing the initial leverage.

For every feasible h_0 , the objective-independent model output is the evaluation map

$$h_0 \mapsto \left(p_{\text{liq}}(T, h_0), M_1(T, h_0), \dots, M_K(T, h_0), \frac{M_1(T, h_0)}{p_{\text{surv}}(T, h_0)}, \dots, \frac{M_K(T, h_0)}{p_{\text{surv}}(T, h_0)} \right), \quad (6)$$

where $M_k(T, h_0) = \mathbb{E}[\Pi_T^k]$ and K is any fixed admissible integer order. Sections 4–5 construct this map. A decision maker may subsequently apply a liquidation limit or utility criterion to it; alternative examples are kept separate in Section 7.

3. Bivariate Kou Dynamics and Exponential Tilting

3.1. Bivariate Kou model

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$dX_{i,t} = \mu_i^X dt + \sigma_i dB_{i,t} + d\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \quad i = 1, 2, \quad (7)$$

where B_1, B_2 are correlated standard Brownian motions with $d\langle B_1, B_2 \rangle_t = \rho dt$, N_i is a Poisson process with intensity λ_i independent of (B_1, B_2) , and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \quad \varphi_{J_i}(u) = \frac{p_i}{1-iu\eta_{i,+}} + \frac{1-p_i}{1+iu\eta_{i,-}}, \quad (8)$$

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1 - p_i}{1 + s\eta_{i,-}}, \quad s \in \left(-\frac{1}{\eta_{i,-}}, \frac{1}{\eta_{i,+}}\right). \quad (9)$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator χ_i defined below to be finite.

The parameter μ_i is the *continuously compounded expected asset-price growth rate*, i.e. $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1 - \eta_{i,+}} + \frac{1 - p_i}{1 + \eta_{i,-}} - 1. \quad (10)$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^X + \frac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching to $e^{\mu_i t}$ gives the log-price drift coefficient

$$\mu_i^X := \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \quad (11)$$

All analytical and numerical objects in the sequel use μ_i^X .

The bivariate process (X_1, X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned} \Psi(u, v) := & i(\mu_1^X u + \mu_2^X v) - \frac{1}{2}(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv) \\ & + \lambda_1(\varphi_{J_1}(u) - 1) + \lambda_2(\varphi_{J_2}(v) - 1), \end{aligned} \quad (12)$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t} + vX_{2,t})}] = e^{t\Psi(u,v)}$.

Table 1 collects all model parameters and their roles.

3.2. Tilted dynamics

For each integer $k \geq 0$, define the k -tilted measure $\mathbb{P}^{(k)}$ by

$$\left. \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp(kX_{2,t} - t\Psi(0, -ik)), \quad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$). *Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)}s + \frac{1}{2}\sigma_Z^2 s^2 + \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - M_2(k)), \quad (13)$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \quad \mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies

$$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}. \quad (14)$$

▷ *Proof, p. 29*

This eigenrelation is the key identity used in Section 5 to turn the Laplace-domain problem into a finite exponential representation.

Remark 3 (Superscript (k) convention). A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untitled case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$. For $k \geq 1$, the k -th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$. The constraint $k\eta_{2,+} < 1$ makes the tilted measure well-defined, while the payoff forcing

additionally requires $k\eta_{1,+} < 1$ so that the long-leg exponential moments $M_1(j)$ are finite for $j = 0, \dots, k$. Thus a payoff-moment order is *admissible* when

$$k\eta_{1,+} < 1, \quad k\eta_{2,+} < 1. \quad (15)$$

Throughout, k is an integer. There is no restriction to $k = 1$ or $k = 2$: every fixed positive integer satisfying (15) is covered.

For the Kou structure, the tilt transforms the jump rates of Z into the k -dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z : the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$. The rates $r_{1,-}^{(k)}$ and $r_{2,-}^{(k)}$ enter the finite-dimensional system in Section 5. The associated phase operators and their truncation at the liquidation boundary are developed separately in Appendix B.

Table 2 collects the quantities introduced in this subsection.

4. Reduction of Killed Payoff Moments

Fix any integer $k \geq 1$ satisfying the two moment-domain conditions (15). Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is intractable: the payoff couples the joint process (X_1, X_2) to the barrier event $\{\tau > T\}$ in a way that is neither separable nor analytically tractable. An exponential change of measure resolves this by tilting the path measure by $e^{kX_{2,T}}$, which absorbs the short-leg factor into the density and leaves a purely one-dimensional killed expectation for the log-ratio $Z = X_1 - X_2$ under the k -tilted measure $\mathbb{P}^{(k)}$. Since Z remains a Lévy process under $\mathbb{P}^{(k)}$, no information about the joint dynamics is lost: the tilted characteristic exponent encodes the full marginal law of Z . The k -th moment then factorises as the known scalar $e^{T\Psi(0, -ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ times the barrier-killed expectation $m_k(T; h_0)$ under $\mathbb{P}^{(k)}$, which is computed via the Laplace–resolvent of Section 5.

The argument below is order-generic and is applied independently at each desired integer k . With distinct downward phase rates, the resulting resolvent has a 3×3 barrier system; coincident rates reduce it to 2×2 . Higher integer orders enlarge only the finite forcing sum from $j = 0, \dots, k$ and do not enlarge either system.

Define

$$g_k(z; h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$, so $g_k(Z_T; h_0) = (e^{h_0+Z_T} - b)^k > 0$ and is a polynomial in e^{Z_T} . From (2),

$$\begin{aligned} \Pi_T^k &= \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}} g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}. \\ \mathbb{E}_{\mathbb{P}}[\Pi_T^k] &= \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{T\Psi(0, -ik)} \mathbb{E}_{\mathbb{P}^{(k)}}[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}]. \end{aligned} \quad (16)$$

Proposition 4.1 (Every admissible integer-order moment). *Let $k \geq 1$ be an integer satisfying (15). Define*

$$m_k(T; h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}].$$

Then

$$\mathbb{E}[\Pi_T^k] := \mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0, -ik)}}{(e^{h_0} - b)^k} m_k(T; h_0),$$

where $m_k(T; h_0)$ is computed via the Laplace–resolvent method of Section 5. Consequently, the method computes $\mathbb{E}[\Pi_T^k]$ for every integer

$$k \in \mathcal{K} := \{n \in \mathbb{N} : n\eta_{1,+} < 1 \text{ and } n\eta_{2,+} < 1\}.$$

Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau > T\}}$, the conditional k -th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{\mathbb{E}[\Pi_T^k]}{p_{\text{surv}}(h_0)}, \quad k \geq 1, \quad k\eta_{1,+} < 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}, \quad \text{Var}(\Pi_T \mid \tau > T) = \frac{\mathbb{E}[\Pi_T^2]}{p_{\text{surv}}(h_0)} - \left(\frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)} \right)^2.$$

The same killed moments also give the unconditional variance of the zero-after-liquidation benchmark,

$$\text{Var}(\Pi_T) = \mathbb{E}[\Pi_T^2] - \mathbb{E}[\Pi_T]^2.$$

Thus survival, killed moments, unconditional moments, and conditional surviving-path moments are all outputs of the same construction. No preference parameter enters this reduction; Section 7 later combines these outputs under several illustrative decision rules.

Remark 4 (No-shorting limit). As $h_0 \rightarrow \infty$ the holding map gives $n_2 S_{2,0} \rightarrow 0$ and $n_1 S_{1,0} \rightarrow 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every admissible integer $k \geq 1$,

$$\lim_{h_0 \rightarrow \infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik, 0)},$$

which provides a useful diagnostic: the formula for $\mathbb{E}[\Pi_T^k]$ must satisfy this limit exactly. The $k = 1$ case gives $e^{\mu_1 T}$ by the drift normalisation (11).

The remainder of the analysis is devoted to computing $m_k(T; h_0)$ for every admissible integer $k \geq 1$ and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

5. Laplace Transforms of the Killed Moments

This section computes the time-Laplace transform of the reduced killed moment $m_k(T; h_0)$ from Proposition 4.1. We first express that target as a killed resolvent, then evaluate it through a fixed-dimensional linear system. The lower-boundary calculation specific to downward jumps is isolated in Appendix B.

5.1. Target Transform and Resolvent

Fix an integer $k \geq 0$ satisfying (15) (automatic at $k = 0$), and define

$$G_k(x) := (e^x - b)^k = \sum_{j=0}^k d_{k,j} e^{jx}, \quad d_{k,j} := \binom{k}{j} (-b)^{k-j}. \quad (17)$$

For a starting distance $x > 0$ above the translated barrier, let

$$\tau_0^{(x)} := \inf\{s \geq 0 : x + Z_s \leq 0\}, \quad \tilde{U}_k(t, x) := \mathbb{E}_{\mathbb{P}^{(k)}} \left[G_k(x + Z_t) \mathbf{1}_{\{\tau_0^{(x)} > t\}} \right].$$

In the non-degenerate diffusion regime of Assumption 5.1, this entrance time agrees almost surely with the strict-breach convention $x + Z_s < 0$ used in Section 2. Then $m_k(t; h_0) = \tilde{U}_k(t, h_0)$ for $k \geq 1$. For uniform notation, set $p_{\text{surv}}(t, h_0) := \mathbb{P}(\tau > t)$ and $m_0(t; h_0) := p_{\text{surv}}(t, h_0)$. Since $G_0 \equiv 1$, the same identity $m_k(t; h_0) = \tilde{U}_k(t, h_0)$ then holds for every $k \geq 0$ considered below.

For q to the right of the time-Laplace abscissa of convergence, define

$$\hat{U}_k(q, x) := \int_0^\infty e^{-qt} \tilde{U}_k(t, x) dt, \quad \hat{m}_k(q; h_0) := \int_0^\infty e^{-qt} m_k(t; h_0) dt = \hat{U}_k(q, h_0). \quad (18)$$

At $k = 0$, (18) is the Laplace transform of the survival probability.

We henceforth regard $\tilde{U}_k(t, \cdot)$ as a function on $\mathbb{R}$ by setting it equal to zero on $(-\infty, 0]$. With killing encoded by this convention, no second generator is needed, and

$$\partial_t \tilde{U}_k(t, x) = \mathcal{L}^{(k)} [\tilde{U}_k(t, \cdot)](x), \quad x > 0, \quad (19)$$

with initial condition $\tilde{U}_k(0, x) = G_k(x)$ for $x > 0$. Integrating (19) by parts in t gives

$$q \hat{U}_k(q, x) - G_k(x) = \mathcal{L}^{(k)} [\hat{U}_k(q, \cdot)](x).$$

Equivalently, under the zero-extension convention, the target solves

$$(q - \mathcal{L}^{(k)}) \hat{U}_k(q, x) = G_k(x), \quad x > 0, \quad \hat{U}_k(q, x) = 0 \quad (x \leq 0). \quad (20)$$

Appendix B verifies how the zero extension acts on the jump terms and derives the boundary equations used below.

5.2. Time-Laplace Transform and Finite-Dimensional System

We solve (20) under the following regularity assumption.

Assumption 5.1 (Generic six-root regime). Fix an integer $k \geq 0$ satisfying (15). Let real q lie strictly to the right of the relevant time-Laplace abscissa of convergence. Assume:

- (i) $\sigma_Z > 0$;
- (ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1^{(k)}(q), \dots, \gamma_6^{(k)}(q)$, ordered so that

$$\text{Re}(\gamma_{1,2,3}^{(k)}) > 0, \quad \text{Re}(\gamma_{4,5,6}^{(k)}) < 0;$$

- (iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \dots, k$.

The forcing expansion (17) and generator eigenrelation (14) motivate the interior ansatz

$$\hat{U}_k(q, x) = \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jx} + \sum_{m=1}^3 B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)x}, \quad x > 0. \quad (21)$$

The first sum is the particular response to the forcing G_k . The second is a homogeneous barrier correction. As the starting distance $x \rightarrow +\infty$, the effect of the lower barrier must vanish and the killed resolvent must approach its unrestricted counterpart. A term $e^{\gamma x}$ with $\text{Re}(\gamma) > 0$ would instead diverge, so the three positive-real-part roots are inadmissible. The ansatz therefore retains only the three roots with negative real part, whose exponentials decay. Their coefficients $B_m^{(k)}(q)$ remain to be determined. The following proposition states the desired transform; Appendix B supplies the downward-phase boundary calculation behind its three matrix rows.

Proposition 5.1 (Laplace-transformed killed moments). *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \quad (22)$$

the coefficients $\mathbf{B}^{(k)}(q) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ are the unique solution of

$$\mathbf{M}_{\text{bar}}^{(k)}(q) \mathbf{B}^{(k)}(q) = -\mathbf{b}^{(k)}(q), \quad (23)$$

where

$$\mathbf{M}_{\text{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4^{(k)}(q)} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5^{(k)}(q)} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6^{(k)}(q)} \\ \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4^{(k)}(q)} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5^{(k)}(q)} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6^{(k)}(q)} \end{pmatrix} \quad (24)$$

and

$$\mathbf{b}^{(k)}(q) = \begin{pmatrix} \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \\ \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + j} \\ \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + j} \end{pmatrix}. \quad (25)$$

The matrix (24) is invertible, so

$$\mathbf{B}^{(k)}(q) = -(\mathbf{M}_{\text{bar}}^{(k)}(q))^{-1} \mathbf{b}^{(k)}(q).$$

The Laplace-transformed killed moment is therefore

$$\widehat{m}_k(q; h_0) = \widehat{U}_k(q, h_0) = \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jh_0} + \sum_{m=1}^3 B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)h_0}. \quad (26)$$

▷ *Proof, p. 29*

The generic system (23) assumes distinct downward rates. If the two rates coincide, they form one phase and the reduced construction is 2×2 ; the details are given in Appendix B, Remark 6.

5.3. Moment Recovery and Survival

Numerical Laplace inversion of (26) recovers the time-domain killed moment,

$$m_k(T; h_0) = \mathcal{L}_{q \rightarrow T}^{-1}[\widehat{m}_k(q; h_0)]. \quad (27)$$

The survival probability is the same construction at $k = 0$. Since $\psi_Z^{(0)}(0) = 0$, one has $P_0(q, x) = 1/q$ and, when $\eta_{1,-} \neq \eta_{2,+}$,

$$\widehat{p}_{\text{surv}}(q; h_0) = \widehat{U}_0(q, h_0) = \frac{1}{q} + \sum_{m=1}^3 B_m^{(0)}(q) e^{\gamma_{m+3}^{(0)}(q)h_0}, \quad \mathbf{M}_{\text{bar}}^{(0)}(q) \mathbf{B}^{(0)}(q) = - \begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix}. \quad (28)$$

Hence

$$p_{\text{surv}}(T, h_0) = \mathcal{L}_{q \rightarrow T}^{-1}[\hat{p}_{\text{surv}}(q; h_0)], \quad p_{\text{liq}}(T, h_0) = 1 - p_{\text{surv}}(T, h_0).$$

If the two untilted downward rates coincide, the $k = 0$ instance of Appendix B, Remark 6, gives the corresponding 2×2 system.

6. Numerical Implementation and Validation

6.1. Validation design and empirical parameters

We validate survival ($k = 0$) and killed moments of orders $k = 1, 2, 3, 4$, examine Talbot-order and Monte Carlo monitoring convergence, and verify the analytical no-shorting limit as $h_0 \rightarrow \infty$. The moment representation is defined for any fixed integer order satisfying the two tail conditions in (15); the numerical evidence in this section is deliberately limited to orders zero through four. We make no numerical stability claim for all higher orders allowed by the formal moment domain.

Time-domain quantities are recovered with the Talbot deformed-contour algorithm (Talbot, 1979). The resolvent is derived first to the right of its Laplace abscissa and then continued to nonsingular contour nodes. The Monte Carlo comparison uses a finite monitoring grid, so it is a discretization benchmark rather than a continuous-time ground truth.

Remark 5 (Continuation to a Talbot contour). Talbot inversion evaluates (26) at complex q reached by analytic continuation. A valid contour deformation avoids transform poles, phase poles, and root collisions, continuing each characteristic-root branch from its right-half-plane value at real q . The sign classification in Assumption 5.1 is therefore an initial labeling, not a fresh test at every complex contour node.

The Kou model and contract parameters are given in Table 3. Parameters are calibrated by the penalized ECF criterion in Appendix A using 540 four-hour WBTC (asset 1) and WETH (asset 2) log-returns on AAVE v3 Ethereum after subtracting an exponentially weighted return mean; the drifts μ_i are adjusted for AAVE supply/borrow APRs (see caption). The AAVE liquidation threshold $b = 0.78$ and maximum origination LTV 0.73 impose a minimum feasible log-buffer $h_0^{\min} = \log(0.78/0.73) \approx 0.066$. WBTC and WETH have distinct downward phase rates for the validated orders $k = 0, \dots, 4$, so the generic barrier system applies in the reported calculations. Because the first moment is weakly identified in high-frequency ECF matching, the fitted μ_i values in Table 3 should be read as model price-growth normalizers rather than directional forecasts; user views can replace them in downstream moment and liquidation calculations.

6.2. Survival, killed moments, and the no-shorting limit

For each initial health factor $H_0 = e^{h_0}$, with $h_0 \in \{h_0^{\min}, 0.10, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00\}$ and $h_0^{\min} = 0.066249$, we report the survival probability p_{surv} and the following conditional statistics of $\Pi_T^k \mid \{\tau > T\}$, where $\mu_k := \mathbb{E}[\Pi_T^k \mid \tau > T]$:

$$\begin{aligned} \text{mean} &:= \mu_1, & \text{variance} &:= \mu_2 - \mu_1^2, \\ \text{skewness} &:= \frac{\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3}{(\mu_2 - \mu_1^2)^{3/2}}, & \text{ex. kurt} &:= \frac{\mu_4 - 4\mu_1\mu_3 + 6\mu_1^2\mu_2 - 3\mu_1^4}{(\mu_2 - \mu_1^2)^2} - 3. \end{aligned} \quad (29)$$

Table 4 reports the objective-independent quantities that are the paper's primary numerical output. The identity $M_k = p_{\text{surv}} \mathbb{E}[\Pi_T^k \mid \tau > T]$ also provides a direct implementation check at each buffer.

As $h_0 \rightarrow \infty$ the short holding $n_2 \rightarrow 0$ and the portfolio converges to a unit long position in asset 1 (Remark 4), so all conditional statistics converge to those of $e^{X_{1,T}}$ under $\mathbb{P}$:

$$\lim_{h_0 \rightarrow \infty} \mu_k = \mathbb{E}_{\mathbb{P}} \left[e^{kX_{1,T}} \right] = \exp(T\Psi(-ik, 0)).$$

Table 5 reports all statistics together with these analytical limits.

6.3. Monitoring and inversion diagnostics

Table 7 isolates monitoring error at $h_0 = 0.10$. Refining the Euler–Poisson grid moves the MC survival estimate toward the Laplace value 0.461481; at 1008 intervals the latter lies inside the reported binomial interval. This does not turn the simulation into exact continuous monitoring—jump ordering and Brownian crossings within an interval remain approximated—but it supports the interpretation of the coarse-grid discrepancy in Table 6.

For the feasible lower buffer $h_0 = 0.066249$, changing the Talbot order from 12 to 40 changes p_{surv} and M_1, M_2 only beyond the eighth reported decimal; the largest relative change between $M = 32$ and $M = 40$ among $k = 0, \dots, 4$ is 4.0×10^{-8} (for M_4). Over the $M = 32$ contour nodes and the same orders, the maximum relative characteristic-root residual is 1.4×10^{-13} , the minimum root-to-pole distance is 3.51, and the largest barrier-matrix condition number is 23.54. The maximum linear-system residual is 3.6×10^{-18} .

These quantities are reported by `jobs/numerical_diagnostics.py`. The current implementation selects the three root branches by real-part rank at each contour node rather than by a global nearest-branch continuation algorithm. The residual, separation, pole, and conditioning checks support that selection for this calibration and $k = 0, \dots, 4$; they do not justify extrapolation to every formally admissible higher order.

6.4. Parameter sensitivity

Figure 1 shows how p_{surv} (left column) and the conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$ (right column) vary with H_0 as each parameter is swept in turn. For the liquidation threshold b (row e–f), survival probability is independent of b at fixed H_0 — only the payoff kernel changes — so the left panel shows the initial leverage $L_0 = H_0/(H_0 - b)$ instead.

Three patterns dominate. (i) *Volatility* is the strongest driver: at $0.3\times$ scale, almost all paths survive for $H_0 \gtrsim 1.35$; at $2\times$ scale, the same cushion yields survival below 40%. (ii) *Correlation* acts through the effective variance of $Z = X_1 - X_2$: the calibrated benchmark already sits at $\rho = 0.858$, where the spread variance is small; moving to lower ρ sharply depresses the survival curve, while its effect on the plotted conditional mean is small for this calibration. Thus the numerical impact is primarily reflected in survival here, without implying a general separation. (iii) *Jump intensity* and *horizon* compress survival monotonically, while increasing the conditional mean at low H_0 through selective survivor effects. For b , initial leverage $L_0 = H_0/(H_0 - b)$ reaches $\approx 6.3\times$ at the AAVE minimum $H_0^{\min} \approx 1.068$ for $b = 0.90$, explaining the strong sensitivity of the conditional mean to b near the minimum buffer.

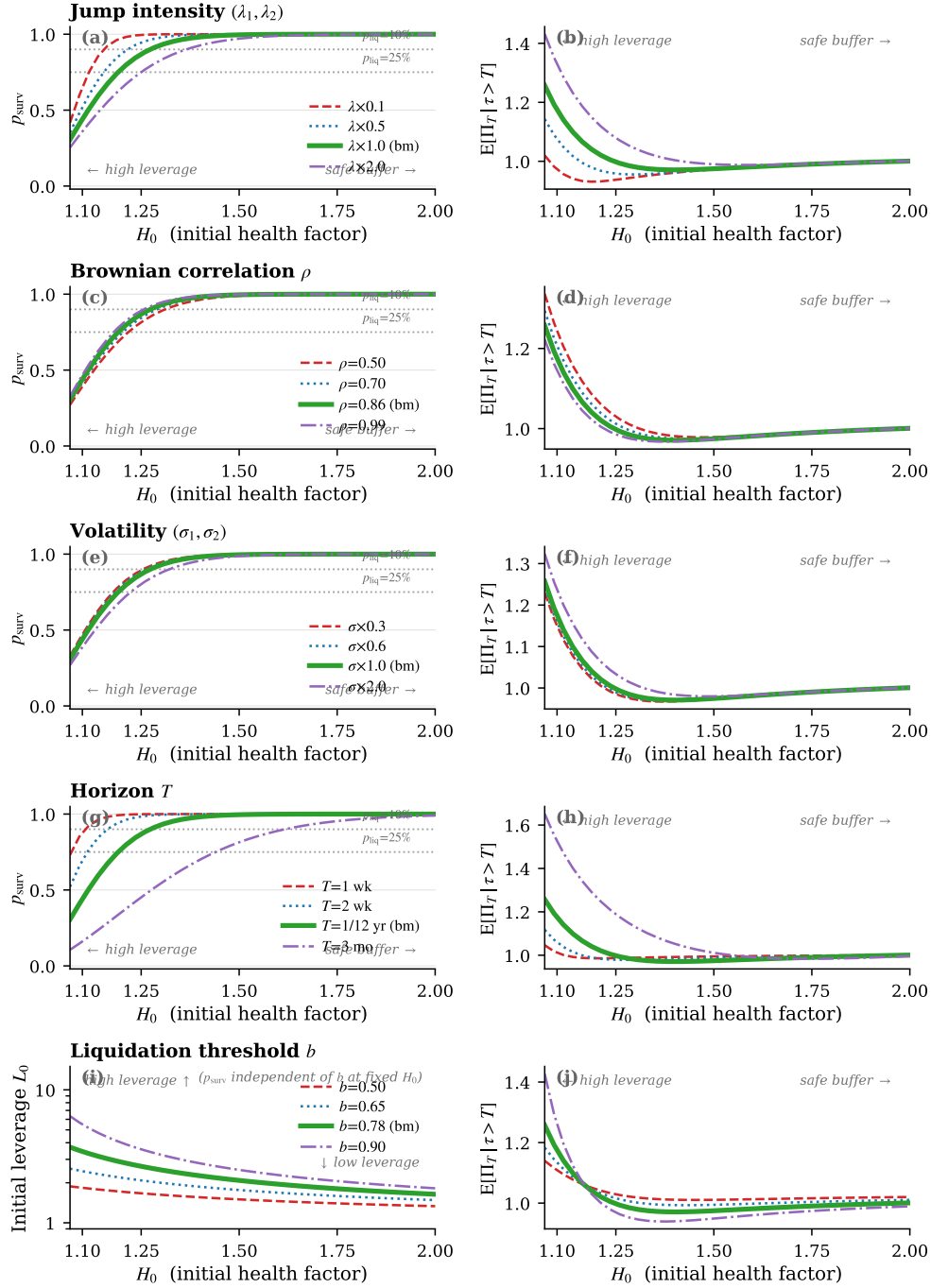


Figure 1: Left column: survival probability p_{surv} (rows a–d) and initial leverage L_0 (row e, log scale) versus H_0 . Right column: conditional mean $\mathbb{E}[\Pi_T | \tau > T]$. Each row varies one parameter; all others fixed at Table 3. Encoding: colour and line style identify parameter values (solid/green = benchmark); the heavier line is the baseline. Laplace–resolvent, $H_0 \in [1.068, 2.000]$, 40 points per curve.

Symbol	Asset	Economic meaning / role	Constraint
μ_i	$i = 1, 2$	Continuously compounded expected price growth rate: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$	
σ_i	$i = 1, 2$	Diffusion volatility of log-price X_i	$\sigma_i > 0$
ρ	—	Correlation between the two Brownian drivers	$ \rho < 1$
λ_i	$i = 1, 2$	Jump arrival intensity (expected jumps per unit time)	$\lambda_i > 0$
p_i	$i = 1, 2$	Probability that a jump in X_i is upward	$p_i \in (0, 1)$
$\eta_{i,+}$	$i = 1, 2$	Mean size of upward jumps in X_i	$0 < \eta_{i,+} < 1$
$\eta_{i,-}$	$i = 1, 2$	Mean size of downward jumps in X_i	$\eta_{i,-} > 0$
χ_i	$i = 1, 2$	Price-jump compensator $M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1]$; enters μ_i^X via (11)	
μ_i^X	$i = 1, 2$	Actual drift coefficient in the log-price SDE (7)	
$\Psi(u, v)$	—	Joint characteristic exponent; $e^{T\Psi(0, -ik)} = \mathbb{E}[e^{kX_{2,T}}]$ appears in the moment formula	
$S_{i,0}$	$i = 1, 2$	Initial price of asset i	$S_{i,0} > 0$
b	—	Liquidation threshold: maximum borrow-to-collateral ratio V_2/V_1 before forced liquidation (Aave v3 liquidation threshold; distinct from maximum LTV)	$b \in (0, 1)$
$\ell_{\max}$	—	Optional maximum loan-to-value ratio $V_{2,0}/V_{1,0}$ permitted at origination	$0 < \ell_{\max} < b$
T	—	Investment horizon	$T > 0$
n_i	$i = 1, 2$	Number of coins held: $n_1 > 0$ long in S_1 , $n_2 > 0$ short in S_2	$n_i > 0$
H_t	—	Health factor: $b n_1 S_{1,t} / (n_2 S_{2,t})$; liquidated when $H_t < 1$	
h_t	—	Log health factor: $h_t = \log H_t = h_0 + Z_t$	
h_0	—	Initial log health factor; evaluation and sizing parameter; determines n_i via (3)	$h_0 \in \mathcal{H}(b, \ell_{\max})$
τ	—	Liquidation time: $\tau = \inf\{t \geq 0 : H_t < 1\}$	
Π_T	—	Killed terminal payoff: $(n_1 S_{1,T} - n_2 S_{2,T}) \mathbf{1}_{\{\tau > T\}}$	

Table 1: Model parameters and derived quantities. The first block contains freely chosen Kou model inputs; the second block contains derived model quantities; the third block contains portfolio and contract inputs; the fourth block contains derived portfolio quantities.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$d\mathbb{P}^{(k)} / d\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Non-negative integer; (15) for $k \geq 1$	Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z ; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathcal{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathcal{L}^{(k)}e^{sz} = \psi_Z^{(k)}(s)e^{sz}$; killing is imposed by zero extension in Section 5
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$

Table 2: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

Group	Parameter	WBTC	WETH
Kou	ECF price-growth normalizer μ_i	0.487	1.200
	Log-price drift μ_i^X	-0.305	0.257
	Model log-return drift $\mathbb{E}[dX_i]/dt$	0.425	1.111
	Empirical de-meaned log drift	0.352	0.606
	Diffusion volatility σ_i	0.241	0.307
	Jump intensity λ_i	372.95	588.95
	Upward-jump probability p_i	0.508	0.357
	Mean upward-jump size $\eta_{i,+}$	0.0111	0.0125
	Mean downward-jump size $\eta_{i,-}$	0.00743	0.00469
Correlation / contract	Brownian correlation ρ	0.858	
	Liquidation threshold b	0.78	
Portfolio	Initial price $S_{i,0}$	75,060.57	2,060.33
	Horizon T	1/12 (one month)	

Table 3: Kou model and contract parameters calibrated by penalized ECF matching to four-hour WBTC/WETH log-returns on AAVE v3 Ethereum (2026-02-26 – 2026-05-27), after EWM de-meaning with span 30. Drifts adjusted for AAVE average APRs: $\mu_{1,+} = 0.0056\% \text{ yr}^{-1}$ (WBTC supply), $\mu_{2,+} = 2.7855\% \text{ yr}^{-1}$ (WETH borrow). Jump intensities are estimated under the broad calibration bound $\lambda_{\max} = 5000$. AAVE liquidation threshold $b = 0.78$; maximum origination LTV 0.73 gives $h_0^{\min} \approx 0.066$. Derived quantities: $\lambda_1\chi_1 + \frac{1}{2}\sigma_1^2 \approx 0.793$, $\lambda_2\chi_2 + \frac{1}{2}\sigma_2^2 \approx 0.943$, $\mu_1^X \approx -0.305$, $\mu_2^X \approx 0.257$. The empirical de-meaned price-growth normalizers are 0.432 (WBTC) and 0.755 (WETH), so the fitted WETH μ_2 is a weakly identified ECF normalizer rather than a robust return forecast. Initial prices are the last aligned AAVE sample, at block 25,189,091.

h_0	p_{surv}	M_1	M_2	$\mathbb{E}[\Pi_T \mid \tau > T]$	$\mathbb{E}[\Pi_T^2 \mid \tau > T]$
0.0662	0.306293	0.385094	0.507211	1.257274	1.655967
0.1000	0.461481	0.536822	0.655481	1.163260	1.420385
0.2500	0.912938	0.898984	0.934991	0.984716	1.024156
0.5000	0.999613	0.984080	1.003811	0.984461	1.004199
1.0000	1.000000	1.015788	1.050402	1.015788	1.050402
2.0000	1.000000	1.033907	1.081910	1.033907	1.081910

Table 4: Laplace–resolvent survival probabilities, killed raw moments, and conditional raw moments for selected feasible buffers under Table 3. The first row uses the protocol origination bound $h_0^{\min} = \log(0.78/0.73)$. Values are generated by `h0_liquidation_moment_report`, which exposes killed and conditional outputs separately.

h_0	H_0	p_{surv}		Mean	Variance	Skewness	Ex. Kurt.
		Lap.	MC				
0.066	1.068	0.3063	0.3171	1.2573	0.0752	0.8766	1.0991
0.10	1.11	0.4615	0.4750	1.1633	0.0672	0.8525	1.0083
0.25	1.28	0.9129	0.9156	0.9847	0.0545	0.5676	0.3668
0.50	1.65	0.9996	0.9995	0.9845	0.0350	0.3132	0.3565
0.75	2.12	1.0000	1.0000	1.0043	0.0236	0.3791	0.4093
1.00	2.72	1.0000	1.0000	1.0158	0.0186	0.4158	0.4234
1.50	4.48	1.0000	1.0000	1.0280	0.0145	0.4318	0.4187
2.00	7.39	1.0000	1.0000	1.0339	0.0130	0.4263	0.4017
∞	∞	1.0000	—	1.0414	0.0114	0.4052	0.3612

Table 5: Laplace–resolvent conditional statistics and the no-shorting analytical limit for WBTC/WETH calibrated parameters (Table 3). Mean, variance, skewness, and excess kurtosis are conditional on survival ($\tau > T$) and computed via the resolvent method (Laplace); MC benchmark uses 10^5 Euler–Poisson paths (252 monitoring intervals over the one-month horizon, seed 42). The smallest tabulated buffer is the exact AAVE-feasible minimum ($h_0^{\min} = 0.066249$, $H_0^{\min} = 1.068493$); at this value Laplace gives $p_{\text{surv}} = 0.3063$ versus MC 0.3171 — the 1.08 percentage-point gap reflects discrete-time MC missing intra-period liquidation events at high jump intensity ($\lambda_1 \approx 373$, $\lambda_2 \approx 589$), not path scarcity ($\approx 32,000$ surviving MC paths). The bottom row gives the analytical limits as $h_0 \rightarrow \infty$; Laplace at $h_0 = 100$ matches these values to relative error $< 10^{-6}$, confirming internal consistency of the resolvent construction, Talbot inversion, and the drift normalisation $\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i\chi_i$.

h_0	H_0	Laplace – MC					CPU time (s)	
		Δp_{surv}	ΔMean	$\Delta \text{Var.}$	$\Delta \text{Skew.}$	$\Delta \text{Kurt.}$	Lap.	MC
0.066	1.068	−0.0108	0.0040	−0.0019	−0.0236	−0.1965	0.06	48.67
0.10	1.11	−0.0136	0.0042	−0.0013	−0.0230	−0.1499	0.10	49.15
0.25	1.28	−0.0026	−0.0001	−0.0007	−0.0054	−0.0563	0.10	48.90
0.50	1.65	0.0001	−0.0005	−0.0003	−0.0110	−0.0347	0.10	49.11
0.75	2.12	0.0000	−0.0002	−0.0002	−0.0107	−0.0382	0.10	49.49
1.00	2.72	0.0000	−0.0002	−0.0002	−0.0123	−0.0378	0.10	49.46
1.50	4.48	0.0000	−0.0001	−0.0001	−0.0144	−0.0369	0.10	49.94
2.00	7.39	0.0000	0.0000	−0.0001	−0.0152	−0.0362	0.10	49.76

Table 6: Laplace minus MC differences ($\Delta = \text{Lap.} - \text{MC}$) and wall-clock computation time. The persistent $\Delta p_{\text{surv}} < 0$ reflects the discretely monitored Euler–Poisson scheme (252 intervals over $[0, T]$) missing intra-interval liquidation events; the effect is strongest near $H_0^{\min} \approx 1.068$ where paths frequently approach the barrier. Higher-moment differences shrink as H_0 increases and more paths survive into the interior. The Laplace timing covers five Talbot inversions ($k = 0, \dots, 4$). On the reported host it is roughly 490 times smaller than the particular 10^5 -path MC run; this ratio is hardware-, implementation-, path-count-, and monitoring-grid-dependent, not a universal speed-up constant.

Intervals over $[0, T]$	MC p_{surv}	Binomial s.e.	Approx. 95% interval	Time (s)
63	0.47712	0.00316	[0.47093, 0.48331]	10.54
252	0.47356	0.00316	[0.46737, 0.47975]	12.36
1008	0.46328	0.00315	[0.45710, 0.46946]	13.35

Table 7: Discrete-monitoring convergence for $h_0 = 0.10$, $T = 1/12$, 25,000 paths, and seed 42. The interval is the normal binomial interval for survival incidence and does not include time-discretization uncertainty.

6.5. Calibration and sizing uncertainty

Pointwise parameter sweeps do not quantify joint sampling variation. We therefore apply a paired circular moving-block percentile bootstrap to the 540 four-hour return pairs. Each replicate resamples the same contiguous index blocks for WBTC and WETH, preserving their contemporaneous pairing and short-range ordering, then repeats ECF calibration, the fixed AAVE rate adjustment, the fourth-moment admissibility check, and the downstream sizing rule. The experiment uses 40 replicates, nine-observation blocks (1.5 days), a fixed seed, and two calibration starts per replicate. All 40 calibrations converged and all 40 sizing problems were feasible. Two starts make this a tractable uncertainty illustration; the point calibration itself uses the more extensive 30-start search described in Appendix A.3.

Table 8 propagates each recalibration through the rule that maximizes initial leverage subject to $p_{\text{liq}} \leq 0.10$ on the same 200-point $H_0 \in [1.068, 2.000]$ grid used below. At the point-selected reference buffer $H_0 = 1.274$, survival is 0.902, while its bootstrap interval is wide. The selected buffer and leverage also vary materially across recalibrations. By contrast, selected liquidation probabilities cluster just below 10% because that is the imposed grid constraint, not because liquidation risk is precisely estimated.

Quantity	Point estimate	90% percentile interval
p_{surv} at reference $H_0 = 1.274$	0.902	[0.483, 0.978]
Selected H_0	1.274	[1.175, 1.536]
Selected initial leverage L_0	2.577	[2.032, 2.974]
Selected p_{liq}	0.0980	[0.0943, 0.0997]

Table 8: Paired circular moving-block bootstrap, conditional on the Kou specification, penalized ECF criterion, EWM preprocessing, fixed AAVE rates and protocol terms, chosen block length, and discrete sizing grid. These descriptive intervals measure within-model sampling variability; they are neither model-selection nor out-of-sample forecast intervals. Script `jobs/calibration_uncertainty.py`.

7. Application: Initial Health-Buffer Sizing

The map in (6) can support different initial-buffer rules without changing the underlying moment calculation. This section gives three examples. They are deliberately stylized decision rules, not competing claims about the uniquely correct investor objective.

Write

$$\mu_c(h_0) := \mathbb{E}[\Pi_T \mid \tau > T], \quad v_c(h_0) := \text{Var}(\Pi_T \mid \tau > T),$$

and, under the killed-payoff convention,

$$\mu_u(h_0) := M_1(T, h_0), \quad v_u(h_0) := M_2(T, h_0) - M_1(T, h_0)^2.$$

The subscripts distinguish moments conditional on survival from unconditional moments of the random variable that is set to zero after liquidation.

7.1. Candidate decision rules

Liquidation-constrained rule. Given a surviving-path score J_{surv} and a liquidation budget $\bar{p}$, select an objective-specific buffer from

$$\arg \max_{h_0 \in \mathcal{H}(b, \ell_{\max})} J_{\text{surv}}(h_0) \quad \text{subject to} \quad p_{\text{liq}}(T, h_0) \leq \bar{p}. \quad (30)$$

For example, a policy that seeks the greatest initial exposure allowed by a risk limit sets $J_{\text{surv}} = L_0$.

Unconditional killed-payoff mean–variance rule. For $\alpha \geq 0$, select from

$$\arg \max_{h_0 \in \mathcal{H}(b, \ell_{\max})} \left\{ \mu_u(h_0) - \frac{\alpha}{2} v_u(h_0) \right\}. \quad (31)$$

This rule treats the killed payoff as terminal wealth. Its interpretation therefore depends on the deliberately severe zero-after-liquidation convention and need not coincide with a criterion based on a protocol recovery model.

Conditional performance with an explicit liquidation penalty. For $\alpha, \delta \geq 0$, select from

$$\arg \max_{h_0 \in \mathcal{H}(b, \ell_{\max})} \left\{ \mu_c(h_0) - \frac{\alpha}{2} v_c(h_0) - \delta p_{\text{liq}}(T, h_0) \right\}. \quad (32)$$

Here δ is an explicit preference parameter, not a quantity identified by the price model. Conditional moments can also display survivor selection near the barrier; the penalty controls how the decision rule weighs that feature against liquidation incidence. The paper does not claim that any one of (30)–(32) is universally correct. Their purpose is to show how the same semi-analytic outputs enter different ex-ante preferences.

7.2. Illustrative evaluation frontier and selected buffers

Figure 2 reports the baseline evaluation curve. Its primary output is the mapping from H_0 to conditional mean, variance, liquidation probability, and leverage. The penalized criterion in panel (d) is one marked use of that curve, rather than the paper’s central result. Table 9 compares several rules on the same baseline calibration and makes their preference parameters explicit. A selected value at a candidate-set endpoint is reported as such; it should not be read as an interior or universal optimum.

Rule	Parameters	Selected H_0	Initial L_0	p_{liq}
Maximum leverage under risk limit	$\bar{p} = 0.05$	1.331	2.417	0.0478
Maximum leverage under risk limit	$\bar{p} = 0.10$	1.274	2.577	0.0980
Maximum leverage under risk limit	$\bar{p} = 0.20$	1.214	2.799	0.1976
Unconditional mean–variance	$\alpha = 0.5$	2.000	1.639	$< 10^{-4}$
Unconditional mean–variance	$\alpha = 1$	2.000	1.639	$< 10^{-4}$
Unconditional mean–variance	$\alpha = 5$	2.000	1.639	$< 10^{-4}$
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0)$	1.068	3.704	0.6937
Conditional mean–variance–penalty	$(\alpha, \delta) = (20, 0)$	2.000	1.639	$< 10^{-4}$
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0.25)$	1.068	3.704	0.6937
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0.5)$	2.000	1.639	$< 10^{-4}$

Table 9: Objective-specific selections and preference-parameter sensitivity for the baseline WBTC/WETH illustration over a 200-point grid on $H_0 \in [1.068, 2.000]$. Values are computed from the same survival and moment map; only the decision rule or its stated parameters change. Grid-endpoint selections show why the table is an illustration rather than evidence of a preferred practical buffer. The liquidation-constrained rows maximize L_0 subject to the stated budget.

8. Conclusion and Discussion

We derive an exact Laplace-domain representation and a semi-analytic numerical procedure for survival probabilities and killed payoff moments of a leveraged crypto long–short position whose liquidation boundary is determined by a collateral-to-liability ratio. The initial log health factor reduces the holdings to one parameter, and an exponential measure change reduces each fixed admissible integer-order moment to a killed expectation of the log-price ratio. Under Kou jump–diffusion dynamics and distinct downward phase rates, three barrier conditions give a 3×3 resolvent system.

These objective-independent quantities turn initial-buffer sizing into a one-dimensional optimization: rapid evaluation of survival and killed-moment outputs supports buffer selection under alternative liquidation constraints and moment-based preferences. The numerical results compare the resolvent output with discretely monitored Monte Carlo, verify the analytical no-shorting limit, and illustrate sensitivity across model and contract inputs. The paired block bootstrap further shows that within-model calibration uncertainty can materially shift the selected buffer, so the optimal buffer moves with the objective, its risk parameters, the loss convention, and the calibration itself; no single selection should be read as a universal practical optimum.

The model omits liquidation recovery, dynamic collateral intervention, oracle and keeper delays, stochastic borrowing rates, execution costs, and market impact. Incorporating these features would change investor terminal wealth and may change the selected buffer. The principal output is therefore not a single leverage number, but the map from the initial health buffer to survival probability and payoff-distribution characteristics – the input that any sizing rule, whether the illustrative ones used here or a more complete dynamic control model, ultimately needs to solve for the optimal buffer.

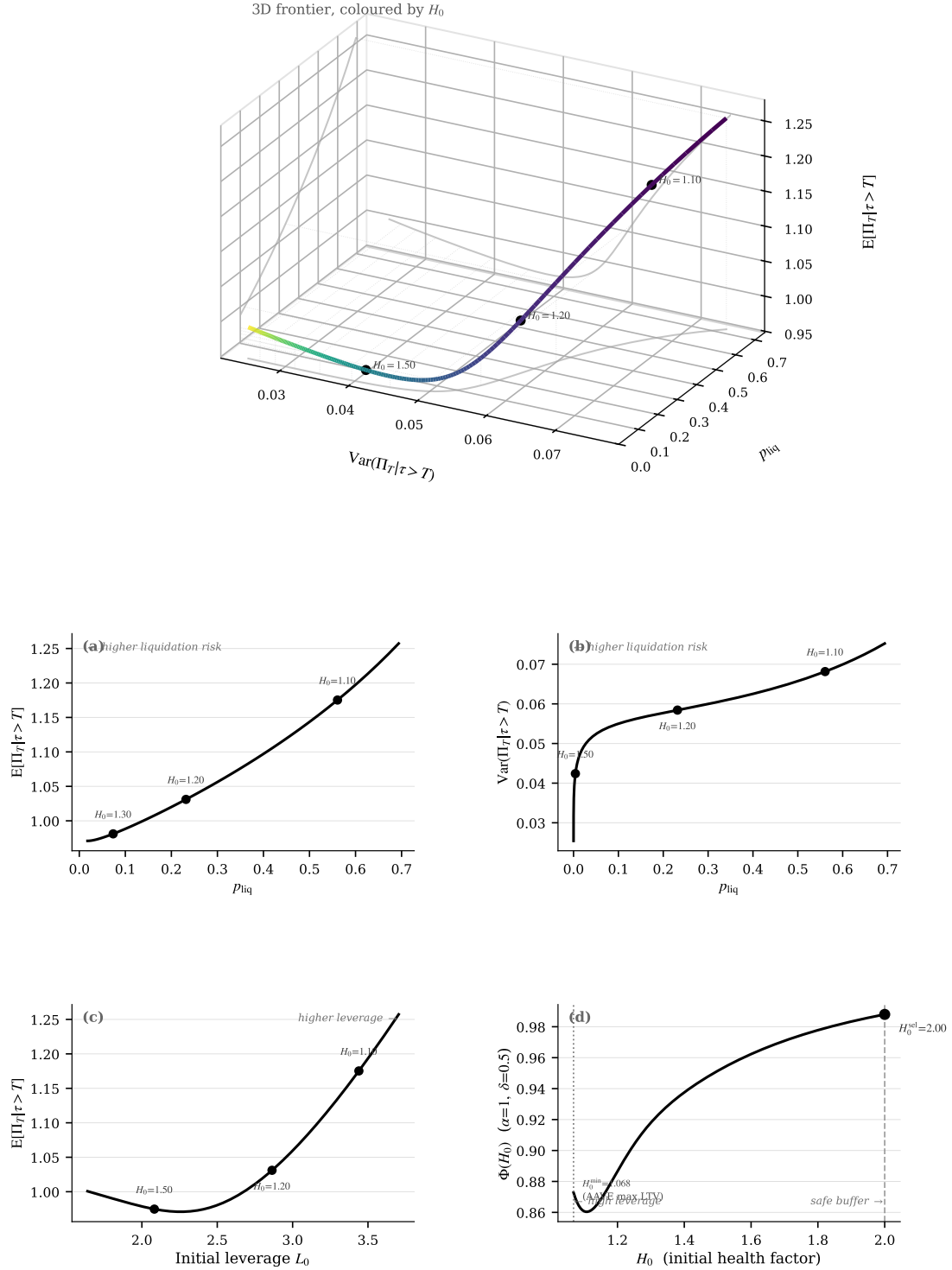


Figure 2: Evaluation map for the benchmark parametrisation. Axes: $x = \text{Var}(\Pi_T | \tau > T)$, $y = p_\ell = 1 - p_{\text{surv}}$, $z = E[\Pi_T | \tau > T]$ (vertical). The curve is parametrised by $H_0 \in [1.068, 2.000]$; the projections show the mean–liquidation, risk–liquidation, and mean–leverage relationships. Panel (d) illustrates the conditional criterion (32) for $(\alpha, \delta) = (1, 0.5)$; it is not designated as the preferred rule. Script `jobs/frontier_analysis.py`.

Appendix A: Empirical Calibration

A.1. Estimator

The bivariate Kou model admits the closed-form Lévy–Khinchine exponent $\Psi(u, v)$ derived in Section 3, so no numerical density inversion is required to compute the model-implied characteristic function

$$\phi_\theta(u, v; \Delta) = \exp(\Delta \Psi_\theta(u, v)), \quad (33)$$

where Δ is the observation interval. Given N observed bivariate log-return pairs $(r_{1,n}, r_{2,n})_{n=1}^N$, the empirical characteristic function is

$$\hat{\phi}_N(u, v) = \frac{1}{N} \sum_{n=1}^N \exp(i(u r_{1,n} + v r_{2,n})). \quad (34)$$

The ECF component of the estimator is the normalized weighted squared distance between (34) and (33) over a discrete frequency grid $\mathcal{G}$:

$$Q_N^{\text{ECF}}(\theta) = \frac{\sum_{(u,v) \in \mathcal{G}} w(u, v) |\hat{\phi}_N(u, v) - \phi_\theta(u, v; \Delta)|^2}{\sum_{(u,v) \in \mathcal{G}} w(u, v)}. \quad (35)$$

Because finite high-frequency samples leave a pronounced intensity–jump-size ridge, the reported calibration uses two transparent soft anchoring terms:

$$\begin{aligned} \hat{\theta} &= \arg \min_{\theta \in \Theta} \mathcal{Q}_N(\theta), \\ \mathcal{Q}_N(\theta) &= Q_N^{\text{ECF}}(\theta) + \omega_\lambda \sum_{i=1}^2 (\log \lambda_i - \log \tilde{\lambda}_i)^2 \\ &\quad + \omega_\eta \sum_{i=1}^2 \sum_{s \in \{+, -\}} (\log \eta_{i,s} - \log \tilde{\eta}_{i,s})^2. \end{aligned} \quad (36)$$

Here $\tilde{\lambda}_i$ is the variance/fourth-cumulant initializer and $\tilde{\eta}_{i,s}$ is the peak-over-threshold (POT) mean-excess initializer described below. The application uses $(\omega_\lambda, \omega_\eta) = (0.05, 0.02)$. These are soft penalties, not equality restrictions; setting both weights to zero recovers pure ECF matching. This is the empirical-characteristic-function approach of Singleton (2001), which is particularly natural for affine and Lévy-based asset-pricing models whose characteristic functions are known in closed form. Feuerverger and McDunnough (1981) establish that ECF estimators over a rich grid can attain maximum-likelihood efficiency. The procedure avoids evaluating the bivariate transition density, which would require a two-dimensional numerical Fourier inversion inside every optimizer call. For background on DEJD likelihood estimation in the univariate case see Ramezani and Zeng (2007).

A.2. Frequency grid

The parameter vector θ has 13 free components; see Table 1. Raw return magnitudes depend on the sampling interval, so fixed raw Fourier frequencies are poorly portable. Let s_1 , s_2 , and s_Z be IQR-based robust scales, $s = \max\{(q_{0.75} - q_{0.25})/1.349, 10^{-8}\}$, for r_1 , r_2 , and $r_1 - r_2$. Dimensionless base frequencies are mapped to raw frequencies by division by the corresponding scale. The grid contains the following groups, with every point accompanied by its conjugate $(-u, -v)$:

- marginal points $(a/s_1, 0)$ and $(0, a/s_2)$ for $a \in \mathcal{A}_{\text{marg}}$;
- same-sign and mixed-sign joint points $(a/s_1, b/s_2)$ and $(a/s_1, -b/s_2)$ for $a, b \in \mathcal{A}_{\text{joint}}$, which are informative about ρ ;
- spread points $(a/s_Z, -a/s_Z)$ for $a \in \mathcal{A}_Z$, which directly match the log-ratio process governing liquidation; and
- high-base marginal and spread points for $a \in \mathcal{A}_{\text{tail}}$, included to retain information about jump-tail shape.

The dimensionless base nodes are

$$\begin{aligned}\mathcal{A}_{\text{marg}} &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10\}, \\ \mathcal{A}_{\text{joint}} &= \{0.25, 0.5, 1, 2, 3, 5\}, \\ \mathcal{A}_Z &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5\}, \\ \mathcal{A}_{\text{tail}} &= \{8, 12, 16, 20, 25\}.\end{aligned}$$

Points whose absolute raw coordinate exceeds 5,000 are discarded to avoid numerical aliasing. Including conjugate pairs, the unfiltered construction has 236 frequency evaluations; this cap leaves 234 for the application sample. Weights depend on dimensionless rather than raw frequency:

$$w(a, b) = \exp\{-0.02(a^2 + b^2)\}.$$

The ordinary spread group receives an additional factor of two. Thus the criterion emphasizes the barrier-driving ratio without making the choice of observation interval determine the effective frequency weights.

A.3. Reparameterisation and initialisation

Unconstrained reparameterisation. Constraints in Table 1 are enforced automatically via the change of variables

$$\sigma_i = e^{a_i}, \quad \lambda_i = e^{b_i}, \quad p_i = \text{sigmoid}(c_i), \quad \eta_{i,+} = \eta_{i,+}^{\max} \text{sigmoid}(d_i), \quad \eta_{i,-} = e^{e_i}, \quad \rho = \tanh(f), \quad (37)$$

where $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$. The unconstrained vector $\tau = (\mu_1, a_1, b_1, c_1, d_1, e_1, \mu_2, \dots, f)$ is optimized with box bounds in this transformed space. For both assets the code uses a scaled sigmoid with

$$\eta_{i,+}^{\max} \leq (1 - \epsilon)/K, \quad i = 1, 2,$$

so that both moment-admissibility conditions in (15) are enforced directly through order K .

Threshold initialiser. The characteristic-function objective is non-linear and non-convex, so the optimisation is initialised by a threshold-based decomposition following the logic of Mancini (2009) for discretely observed jump-diffusions. Let $m_i = \text{median}(r_{i,n})$ and let s_i be the IQR robust scale. For a frequency-aware threshold constant $c \in [3, 5]$, define candidate jump observations

$$\mathcal{J}_i = \{n : |r_{i,n} - m_i| > c s_i\}, \quad \mathcal{C}_i = \{1, \dots, N\} \setminus \mathcal{J}_i.$$

Initial volatility and correlation values use the continuous subset $\mathcal{C}_i$, while the jump-size means are anchored by peak-over-threshold excess means and the intensities are corrected by a variance/fourth-cumulant estimate. Schematically:

$$\begin{aligned} p_i^{(0)} &= \frac{|\mathcal{J}_{i,+}|}{|\mathcal{J}_i|} \quad \text{clamped to the admissible probability range,} \\ \eta_{i,+}^{(0)}, \eta_{i,-}^{(0)} &= \text{POT mean-excess estimates,} \\ \lambda_i^{(0)} &= \text{two-moment estimate from variance and fourth cumulant,} \\ \sigma_i^{(0)} &= \text{sd}(r_{i,n} : n \in \mathcal{C}_i) / \sqrt{\Delta}, & \rho^{(0)} &= \text{corr}(r_1, r_2)_{n \in \mathcal{C}_1 \cap \mathcal{C}_2}. \end{aligned}$$

The initializer first estimates the annualized log-process drift with a damped mean-jump correction,

$$\mu_i^{X,(0)} = \bar{r}_i / \Delta - 0.25 \lambda_i^{(0)} \left[p_i^{(0)} \eta_{i,+}^{(0)} - (1 - p_i^{(0)}) \eta_{i,-}^{(0)} \right],$$

then stores the result in the paper's price-growth convention,

$$\mu_i^{(0)} = \text{clip}_{[\mu_{\min}, \mu_{\max}]} \left(\mu_i^{X,(0)} + \frac{1}{2} (\sigma_i^{(0)})^2 + \lambda_i^{(0)} \chi_i^{(0)} \right).$$

The threshold filter is not the final estimator. It is a warm start that places the optimizer in a sensible region of parameter space; see the remark in [Mancini \(2009\)](#).

Multi-start cloud. Because $\mathcal{Q}_N(\tau)$ may have local minima, the optimizer is run from a structured cloud. It contains the unperturbed initializer and six deterministic intensity–size ridge starts with

$$\lambda_i \mapsto c\lambda_i, \quad \eta_{i,\pm} \mapsto \eta_{i,\pm} / \sqrt{c}, \quad c \in \{0.1, 0.25, 0.5, 2, 4, 8\}.$$

Remaining starts cycle through four high-jump/high-diffusion scenarios and add bounded random perturbations to scale, probability, correlation, and drift parameters. The application calibration uses 30 starts (the library default is 20) and returns the bounded L-BFGS-B minimizer with the lowest penalized objective.

A.4. Identifiability and model-comparison protocol

The penalized criterion $\mathcal{Q}_N(\theta)$ is non-convex. Jump-diffusion models possess approximate ridges along directions that preserve the compound-Poisson variance contribution $\lambda_i \mathbb{E}[J_i^2] = 2\lambda_i(p_i \eta_{i,+}^2 + (1 - p_i) \eta_{i,-}^2)$ ([Ramezani and Zeng, 2007](#)). The diffusion volatilities σ_i and the Brownian correlation ρ are generally better identified from short samples than jump-size and intensity parameters, which carry larger finite-sample uncertainty unless the observation window contains many jump events. Because λ_i is annualized, a one-year window contains λ_i jumps in expectation, irrespective of whether prices are sampled hourly or daily. Finer sampling can make individual jumps easier to separate from aggregated returns, but it does not multiply the underlying annual Poisson count.

For descriptive fit comparisons on a common grid, compare the unpenalized component $\mathcal{Q}_N^{\text{ECF}}(\hat{\theta})$ across nested models (GBM, Merton, bivariate Kou), rather than the penalized values $\mathcal{Q}_N(\hat{\theta})$ whose anchors differ. Such comparisons should be accompanied by liquidation-relevant diagnostics: empirical versus model Z-tail quantiles, empirical versus model barrier-crossing frequencies, and simulated versus observed drawdowns of Z . These are more directly informative for the sizing application than standard likelihood-based criteria. The present numerical illustration does not carry out a formal GBM–Merton–Kou selection exercise. It does propagate within-model sampling variability through the paired circular

moving-block bootstrap reported in Section 6.5. Those intervals remain conditional on the selected Kou model and stated empirical choices; they are not evidence of model dominance or parameter-robust recommendations.

The repository's calibration package contains the implementation. The synthetic calibration job shows parameter recovery and produces Figure 3.

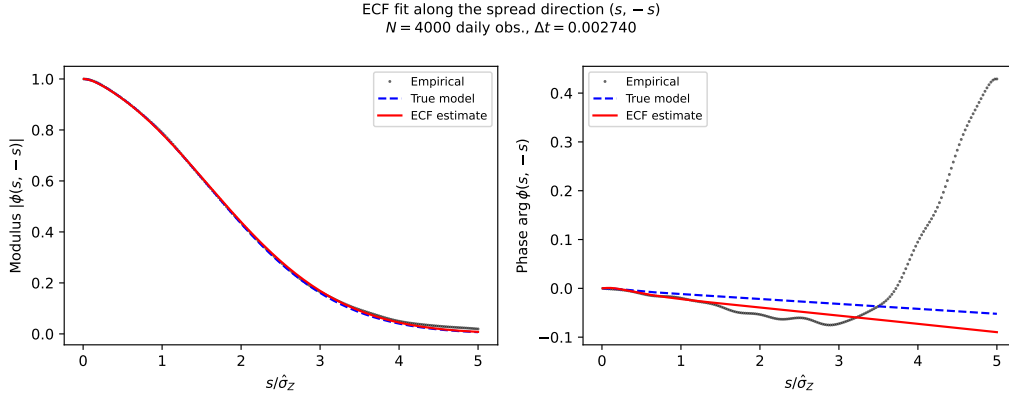


Figure 3: ECF fit along the spread direction $(s, -s)$ for $N = 4,000$ synthetic daily log-return observations. Left: modulus $|\phi(s, -s)|$; right: phase $\arg \phi(s, -s)$. Dots: empirical CF $\hat{\phi}_N$; dashed: true model CF; solid red: ECF estimate. The close agreement in modulus and phase documents the fit along the liquidation-relevant spread direction; this plot alone does not assess every component of the full-grid criterion. Script `jobs/calibrate_kou.py`.

Appendix B: Killing at the Barrier and Downward-Phase Corrections

Section 5 uses only the generator $\mathcal{L}^{(k)}$ from Lemma 3.1, with a killed expectation extended by zero below the barrier. This appendix records how that convention changes the downward jump integrals and derives the three rows of the barrier system (23).

B.1. Phase Operators and Zero Extension

Definition B.1 (Phase operators). For $j = 1, 2$, the phase operators associated with the tilted jump kernels of Z under $\mathbb{P}^{(k)}$ are

$$K_{j,+}^{(k)}[f](x) := \int_0^\infty f(x+y) r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y} dy, \quad K_{j,-}^{(k)}[f](x) := \int_0^\infty f(x-y) r_{j,-}^{(k)} e^{-r_{j,-}^{(k)} y} dy.$$

For any complex γ with $\text{Re}(\gamma) < r_{j,+}^{(k)}$ and $\text{Re}(\gamma) > -r_{j,-}^{(k)}$, direct integration gives

$$K_{j,+}^{(k)}[e^\gamma](x) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma} e^{\gamma x}, \quad K_{j,-}^{(k)}[e^\gamma](x) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma} e^{\gamma x}. \quad (38)$$

This phase construction originates in Kou and Wang (2003) for first-passage problems under the double-exponential model and is used by Sepp (2004) to reduce barrier pricing to a finite linear system.

The tilted phase intensities are

$$\lambda_{1,+}^{(k)} = \lambda_1 p_1, \quad \lambda_{1,-}^{(k)} = \lambda_1 (1 - p_1), \quad \lambda_{2,+}^{(k)} = \frac{\lambda_2 (1 - p_2)}{1 + k\eta_{2,-}}, \quad \lambda_{2,-}^{(k)} = \frac{\lambda_2 p_2}{1 - k\eta_{2,+}}. \quad (39)$$

Using Definition B.1, the full-line action of the generator is

$$\mathcal{L}^{(k)} f(x) = \mu_Z^{(k)} f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^2 \lambda_{a,+}^{(k)} (K_{a,+}^{(k)}[f](x) - f(x)) + \sum_{d=1}^2 \lambda_{d,-}^{(k)} (K_{d,-}^{(k)}[f](x) - f(x)). \quad (40)$$

Applying (38) recovers the exponential symbol (14).

Now let f be defined on the alive half-line $x > 0$ and let $\bar{f} = f \mathbf{1}_{(0,\infty)}$ be its zero extension. Upward jumps remain in the alive region, whereas a downward jump of size $y \geq x$ lands in the absorbing region. Consequently,

$$K_{d,-}^{(k)}[\bar{f}](x) = K_{d,-}^{(k),D}[f](x) := \int_0^x f(x-y) r_{d,-}^{(k)} e^{-r_{d,-}^{(k)} y} dy, \quad x > 0. \quad (41)$$

Substitution into (40) gives

$$\begin{aligned} \mathcal{L}^{(k)}[\bar{f}](x) &= \mu_Z^{(k)} f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^2 \lambda_{a,+}^{(k)} (K_{a,+}^{(k)}[f](x) - f(x)) \\ &\quad + \sum_{d=1}^2 \lambda_{d,-}^{(k)} (K_{d,-}^{(k),D}[f](x) - f(x)), \quad x > 0. \end{aligned} \quad (42)$$

Thus killing changes the arguments on which $\mathcal{L}^{(k)}$ acts, not the identity of the generator. Equation (42) is the standard zero-extension construction for a Lévy process killed at an absorbing boundary.

B.2. Downward-Phase Residual and Barrier Conditions

Lemma B.1 (Truncated downward phase). *For $r > 0$, $\beta \in \mathbb{C} \setminus \{-r\}$, and $x > 0$,*

$$\int_0^x e^{\beta(x-y)} r e^{-ry} dy = \frac{r}{r+\beta} e^{\beta x} - \frac{r}{r+\beta} e^{-rx}. \quad (43)$$

No restriction $\operatorname{Re}(\beta) > -r$ is needed, since the integral is over a finite interval.

▷ *Proof, p. 29*

For a finite exponential sum $v(x) = \sum_{\beta \in \mathcal{E}} a_\beta e^{\beta x}$, define the truncation-correction coefficient

$$\mathfrak{C}_r[v] := \sum_{\beta \in \mathcal{E}} a_\beta \frac{r}{r+\beta}. \quad (44)$$

Then (43) gives

$$K_{r,-}^D[v](x) := \int_0^x v(x-y) r e^{-ry} dy = \sum_{\beta \in \mathcal{E}} a_\beta \frac{r}{r+\beta} e^{\beta x} - \mathfrak{C}_r[v] e^{-rx}. \quad (45)$$

Combining (42) and (45), for exponents at which the upward integrals are finite and which avoid the downward phase poles,

$$\mathcal{L}^{(k)}[\bar{v}](x) = \sum_{\beta \in \mathcal{E}} a_\beta \psi_Z^{(k)}(\beta) e^{\beta x} - \sum_{d=1}^2 \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[v] e^{-r_{d,-}^{(k)} x}, \quad x > 0, \quad (46)$$

where $\psi_Z^{(k)}$ is understood through the meromorphic continuation of its exponential symbol outside the common full-line moment strip.

Apply (46) term by term to the interior ansatz (21). Its forcing coefficients and characteristic-root equations give

$$(q - \mathcal{L}^{(k)}) \widehat{U}_k(q, x) = G_k(x) + \sum_{d=1}^2 \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[\widehat{U}_k(q, \cdot)] e^{-r_{d,-}^{(k)} x}. \quad (47)$$

When the two downward rates are distinct, the residual exponentials are linearly independent, so the interior equation holds if and only if both correction coefficients vanish. Since $\sigma_Z > 0$, the lower boundary is regular for the Brownian component and absorption also imposes the Dirichlet condition. The three conditions are therefore

$$\widehat{U}_k(q, 0) = 0, \quad \mathfrak{C}_{r_{1,-}^{(k)}}[\widehat{U}_k(q, \cdot)] = 0, \quad \mathfrak{C}_{r_{2,-}^{(k)}}[\widehat{U}_k(q, \cdot)] = 0. \quad (48)$$

Substituting (21) into (48) gives

$$\sum_{m=1}^3 B_m^{(k)} = - \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)},$$

and, for $d = 1, 2$,

$$\sum_{m=1}^3 \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + \gamma_{m+3}^{(k)}(q)} B_m^{(k)} = - \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + j}.$$

These are precisely the three rows of (23).

Lemma B.2 (Invertibility of $\mathbf{M}_{\text{bar}}^{(k)}(q)$). *Under Assumption 5.1 and condition (22), the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}(q)$ in (24) is invertible.*

Remark 6 (Coincident downward rates). If $r_{1,-}^{(k)} = r_{2,-}^{(k)} =: r$, the two downward kernels combine into a single phase. The meromorphic symbol must first be reduced by cancelling the repeated phase factor; otherwise clearing denominators introduces a spurious root at the pole $-r$. The reduced characteristic equation has two genuine decaying roots, say $\gamma_4^{(k)}(q), \gamma_5^{(k)}(q)$, and the conditions $\widehat{U}_k(q, 0) = 0$ and $\mathfrak{C}_r[\widehat{U}_k(q, \cdot)] = 0$ give the 2×2 system

$$\begin{pmatrix} 1 & 1 \\ \frac{r}{r + \gamma_4^{(k)}(q)} & \frac{r}{r + \gamma_5^{(k)}(q)} \end{pmatrix} \begin{pmatrix} B_1^{(k)} \\ B_2^{(k)} \end{pmatrix} = - \begin{pmatrix} \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \\ \sum_{j=0}^k \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r}{r + j} \end{pmatrix}.$$

Thus the 3×3 description applies to the generic distinct-rate regime; the coincident-rate case is a genuine lower-dimensional phase model, not a singular generic system.

Appendix C: Proofs

Proof of Lemma 3.1. Step 1: Laplace transform under $\mathbb{P}^{(k)}$. Because $Z_t = X_{1,t} - X_{2,t}$ and the density on $\mathcal{F}_t$ equals $e^{kX_{2,t} - t\Psi(0, -ik)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}[e^{sZ_t}] = e^{-t\Psi(0, -ik)} \mathbb{E}_{\mathbb{P}}[e^{sX_{1,t} + (k-s)X_{2,t}}] = e^{t[\Psi(-is, -i(k-s)) - \Psi(0, -ik)]}, \quad (49)$$

where we used $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t} + \beta X_{2,t}}] = e^{t\Psi(-i\alpha, -i\beta)}$. This identity is first valid for s in the common moment-generating strip. In particular, $k\eta_{2,+} < 1$ is needed at $s = 0$ to define the tilt, and evaluating the payoff forcing at $s = j$, $j = 0, \dots, k$, additionally requires $k\eta_{1,+} < 1$. Since the Radon–Nikodým density is exponential-linear in $(X_{1,t}, X_{2,t})$, Girsanov’s theorem for Lévy processes guarantees that Z remains a Lévy process under $\mathbb{P}^{(k)}$ with exponent

$$\psi_Z^{(k)}(s) = \Psi(-is, -i(k-s)) - \Psi(0, -ik). \quad (50)$$

Step 2: Expand and simplify. Substituting (12) with $(u, v) = (-is, -i(k-s))$, and noting $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$, and $\varphi_{J_i}(-i\cdot) = M_i(\cdot)$:

$$\begin{aligned} \Psi(-is, -i(k-s)) &= \mu_1^X s + \mu_2^X (k-s) + \frac{1}{2}\sigma_1^2 s^2 + \frac{1}{2}\sigma_2^2 (k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) \\ &\quad + \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - 1). \end{aligned} \quad (51)$$

Setting $s = 0$ in (51) gives $\Psi(0, -ik) = \mu_2^X k + \frac{1}{2}\sigma_2^2 k^2 + \lambda_2(M_2(k) - 1)$. Subtracting and collecting terms by type:

Drift. $\mu_1^X s + \mu_2^X (k-s) - \mu_2^X k = (\mu_1^X - \mu_2^X)s$.

Diffusion. Expanding $(k-s)^2 = k^2 - 2ks + s^2$ and $(k-s) = k-s$:

$$\begin{aligned} &\frac{1}{2}\sigma_1^2 s^2 + \frac{1}{2}\sigma_2^2 (k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) - \frac{1}{2}\sigma_2^2 k^2 \\ &= \frac{1}{2}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s = \frac{1}{2}\sigma_Z^2 s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s. \end{aligned}$$

Jumps. $\lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - 1) - \lambda_2(M_2(k) - 1) = \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - M_2(k))$.

Combining, the drift coefficient is $\mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$, which establishes (13). The eigenrelation (14) follows because for any Lévy process with exponent ψ , the generator satisfies $\mathcal{L}e^{sz} = \psi(s)e^{sz}$, as can be seen by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\Psi(s)}$ with respect to t at $t = 0$. $\square$

Proof of Lemma B.1. Direct integration over the finite interval gives

$$\int_0^x e^{\beta(x-y)} r e^{-ry} dy = e^{\beta x} r \int_0^x e^{-(r+\beta)y} dy = e^{\beta x} \frac{r}{r+\beta} (1 - e^{-(r+\beta)x}),$$

which is (43). Since the interval is finite, this calculation is valid for every $\beta \neq -r$ without a full-line integrability condition. $\square$

Proof of Proposition 5.1. For the ansatz (21), the zero-extension identity (47) shows that the interior resolvent equation holds precisely when its two downward-phase correction coefficients vanish. Absorption adds $\widehat{U}_k(q, 0) = 0$, giving the three conditions (48). As shown in Appendix B, substitution of the ansatz produces exactly the system (23) with matrix (24) and forcing vector (25). Lemma B.2 gives the unique coefficient vector.

The resulting zero extension solves (20); uniqueness of the killed resolvent in the relevant exponential-growth class identifies it with $\widehat{U}_k$. Finally, (18) and evaluation at $x = h_0$ give (26). $\square$

Proof of Lemma B.2. For brevity, write $\gamma_m := \gamma_m^{(k)}(q)$. Subtracting row 1 from rows 2 and 3 replaces entry (i, m) for $i = 2, 3$ by

$$-\frac{\gamma_{m+3}}{r_{i-1,-}^{(k)} + \gamma_{m+3}}.$$

Pulling out -1 from each of these two rows (net sign $(-1)^2 = 1$) and then factoring γ_{m+3} from each column gives

$$\det(\mathbf{M}_{\text{bar}}^{(k)}) = \gamma_4 \gamma_5 \gamma_6 \det\left(\frac{1}{x_i + \gamma_{j+3}}\right)_{1 \leq i, j \leq 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i + \gamma_{j+3}}\right) = \frac{\prod_{1 \leq i < i' \leq 3} (x_{i'} - x_i) \cdot \prod_{4 \leq m < m' \leq 6} (\gamma_{m'} - \gamma_m)}{\prod_{i=1}^3 \prod_{m=4}^6 (x_i + \gamma_m)}.$$

Since $x_1 = 0$, the factor $\prod_{m=4}^6 (x_1 + \gamma_m) = \gamma_4 \gamma_5 \gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (22); the γ_m are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. $\square$

The survival probability requires no separate proof: it is the $k = 0$ instance of Proposition 5.1 and Lemma B.2 above. At $k = 0$, the forcing term in (21) is $1/q$ because $\psi_Z^{(0)}(0) = 0$, while (22) becomes $\eta_{1,-} \neq \eta_{2,+}$.

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